How to be Mistaken and Still Happy: Belief-Relative Presuppositions and Factivity Illusions

Maik Thalmann and Andrea Matticchio

1 Introduction

Certain emotive predicates, like English *regret* and *be happy*, are traditionally analyzed as carrying a factive presupposition that their complement is true (Kiparsky and Kiparsky 1970; more recently, Uegaki and Sudo 2019, i.a.),[1] because language users tend to infer that their complements are true when they are exposed to utterances containing such predicates, and this inference passes standard tests for presupposition projection. Consider the example in (1): a context where Aditi does not like linguistics makes the utterance infelicitous, and this effect persists if the sentence is negated.

(1) Skye is happy that Aditi likes linguistics.
$\rightsquigarrow$ Aditi likes linguistics.
  a. # Aditi doesn't like linguistics, but Skye is happy that she does.
  b. # Aditi doesn't like linguistics, and Skye is not happy that she does.

This inference also robustly projects out of conditional antecedents. In this, emotive predicates pattern like hard presupposition triggers, whose presuppositions cannot be easily locally accommodated even in ignorance contexts (Abusch 2002, 2010), as shown in (2). The relevant contrast is established with semi-factive predicates, like verbs of discovery, whose presuppositions are soft and can be locally accommodated in the antecedent of a conditional in similar contexts (Abrusán 2016).

(2) a. # I don't know if Aditi likes linguistics, but if Skye is happy that she does, they'll give her a book.
  b. I don't know if Aditi likes linguistics, but if Skye discovers that she does, they'll give her a book.

The factive inference of emotive predicates shows another hallmark of presuppositions: filtering. In a conditional sentence like (3), the presupposition in the consequent does not project globally, but is filtered out by an antecedent that entails it. Therefore, the sentence overall is compatible with a context in which it is not established whether the complement of the emotive predicate is true.

(3) If Aditi likes linguistics, Taro is happy that she does.
$\not\rightsquigarrow$ Aditi likes linguistics.

Despite this evidence, we will argue that emotive predicates do not come with a factive presupposition that their complement is true. The crucial observation, which has been around since Klein 1975 (also cited in Gazdar 1979, Heim 1992, Egré 2008), is that this inference does not arise in contexts of mistaken belief. When the subject of the emotive predicate is ascribed a relevant mistaken belief state, a sentence with an emotive predicate can be uttered felicitously even though the speaker explicitly denies what would amount to the factive inference using *mistakenly*, as shown in (4).

(4) Skye mistakenly believes that Aditi likes linguistics, and they're happy that she does.
$\not\rightsquigarrow$ Aditi likes linguistics.

In this paper, we argue that a close inspection of emotive predicates motivates certain standard assumptions of Satisfaction Theory (Stalnaker 1973, Karttunen 1974, Heim 1983, 1992). In particular, we endorse a view of presuppositions as conditions that natural language sentences impose

[1] Throughout the paper, we will use the expression *emotive predicates* to refer to predicates that have been traditionally called *emotive factives*. We will ignore other attitude predicates with an emotive component that do not have the same type of presupposition, like *hope* or *fear*.

on contexts, and we reject the idea that presuppositions are instead inferences that are readily available to the speaker's introspection. Instead, separate mechanisms, which presumably operate in the domain of pragmatics, determine what inferences speakers can draw from sentences that carry presuppositions (see also a discussion on this point in Fox 2013, Mandelkern and Rothschild 2018). In particular, we assume that *global accommodation* is the main source of inferences on the side of the hearer that is exposed to a sentence that carries a presupposition; the inferences that global accommodation gives rise to can be conditioned by certain additional pragmatic assumptions.

Our proposal to explain the data in (1) to (3), while rejecting the view that emotive predicates carry a factive presupposition, is based on the interaction between global accommodation and a pragmatic principle that introduces a bias about the relation between the speaker's beliefs and the beliefs of the individuals the discourse is about. This principle and its interaction with global accommodation additionally allow us to give a parallel explanation of other, and more well-known puzzles about presuppositions triggered in attitude contexts. This parallelism suggests that a treatment of these cases based on Satisfaction Theory and local satisfaction in attitude contexts (Heim 1992) is more appealing than views that assume default global projection, such as DRT (Geurts 1999) or Dissatisfaction Theory (Mandelkern 2016), besides yielding a less complex look at the data.

2 The Illusion of Two Presuppositions

Some recent work has assumed that emotive predicates come with two presuppositions that are lexically encoded (Uegaki and Sudo 2019, i.a.). One is the traditional factive presupposition, whereas the other, we will call the *belief-relative* presupposition. The belief-relative presupposition corresponds to the factive presupposition embedded under a modal operator that relativizes it to the beliefs of the attitude holder, and it is logically independent from the factive presupposition.

(5) Skye is happy that Aditi likes linguistics.
Seems to presuppose:
a. Aditi likes linguistics. *(factive presupposition)*
b. Skye believes that Aditi likes linguistics. *(belief-relative presupposition)*

The belief-relative presupposition passes all the tests shown in Section 1: it projects out of entailment-canceling operators, like negation or conditional antecedents, and it can be filtered in canonical ways. The projection tests are replicated in (6), where no mention of the actual truth value of the proposition that *Aditi likes linguistics* is made; (7) tests filtering.

(6) a. # Skye doesn't believe that Aditi likes linguistics, but they're happy that she does.
b. # Skye doesn't believe that Aditi likes linguistics, and they're not happy that she does.
c. # I don't know if Skye believes that Aditi likes linguistics, but if they're happy that she does, they'll give her a book.

(7) If Skye believes that Aditi likes linguistics, they're happy that she does.
$\not\rightsquigarrow$ Skye believes that Aditi likes linguistics.

The presence of the belief-relative presupposition can also be diagnosed from the contrast with factive predicates like *know* that do not have one. We assume that *know* minimally asserts belief of the complement and presupposes its truth in the world of evaluation. Consider the contrast between the reasoning sequences in (8) and (9), such that the first is not valid but the second intuitively is.

(8) a. Eleni knows that the man in the cloak is helping.
b. The man in the cloak is her uncle (but Mary doesn't know).
c. $\not\therefore$ Eleni knows that her uncle is helping.

(9) a. Eleni is happy that the man in the cloak is helping.
b. The man in the cloak is her uncle (but Mary doesn't know).
c. $\therefore$ Eleni is happy that her uncle is helping.

In (8), the conclusion is not valid because it has a salient reading where the definite description *her uncle* is read *de dicto* with respect to the attitude predicate *know*. If Mary is not aware that the man in the cloak is her uncle, she does not know *de dicto* that it is her uncle who is helping: in this case, substitution fails. In (9), a *de dicto* reading of the conclusion is instead not possible in a context where Mary does not know that it is her uncle who is helping: in this context, a *de dicto* reading of the conclusion would lead to a presupposition failure if *be happy* presupposes that the subject believes the complement. Therefore, the belief-relative presupposition rules out a *de dicto* reading of the conclusion as infelicitous; on a *de re* reading, the conclusion is necessarily valid.

The two alleged presuppositions in (5) do not have the same status, though. As shown in Section 1, one can be suspended in contexts of mistaken belief. However, as shown in the minimal contrast in (10), the same does not hold for the belief-relative presupposition. Given the same sentence containing the presupposition trigger *be happy*, a context that satisfies the belief-relative presupposition, but not the factive one, is enough to make the utterance felicitous; instead, a context that satisfies the factive presupposition, but not the belief-relative one, leads to presupposition failure. In other words, only one of the two alleged presuppositions can be suspended, namely the factive one. For the contrast in (10), we rely on the meaning of *mistakenly believe that p*, which entails $\neg p$, and on the presupposition of *doesn't know that p*, which entails p.

(10) a. Skye mistakenly believes that Aditi likes linguistics. They're happy that she does.
b. # Skye doesn't know that Aditi likes linguistics. They're happy that she does.

As said in Section 1, we adopt a standard view of presuppositions as conditions that sentences impose on contexts for felicity. If a sentence S leads to an inference that p, but S can be uttered in a context that does not entail p (or *a fortiori*, in a context that entails $\neg p$, as above), then p is not a presupposition of S. Therefore, according to this definition, the factive inference of emotive predicates is not a presupposition, because emotive predicates can be used felicitously in some contexts where their complement is not true, namely in mistaken belief contexts. Instead, there is no evidence that the same predicates can be used in contexts where the belief-relative presupposition is not met, and we assume that this is indeed a presupposition triggered by these predicates.[2]

Although we reject the idea that the factive inference associated to emotive predicates is a presupposition, we maintain that there is a connection between this inference and the actual presupposition that these predicates have, namely the belief-relative one. In Section 3, this connection allows us to sketch an account of the presupposition-like behavior of this inference, such that it passes the tests for projection and filtering just as well as the belief-relative presupposition. We motivate the connection between the factive inference and the belief-relative presupposition on the basis of the existence of a well-known puzzle for presupposition projection that can receive a parallel account.

(11) a. Skye thinks that Aditi, too, fell.
$\rightsquigarrow$ Skye thinks that someone else fell. *(i-inference)*
$\rightsquigarrow$ Someone else fell. *(e-inference)*
b. Skye mistakenly believes that Eleni fell, and they think that Aditi, too, fell.
$\rightsquigarrow$ Skye thinks that someone else (i.e., Eleni) fell.
$\not\rightsquigarrow$ Someone else fell.

Sudo (2014) addresses the issue of the existence of the two inferences in (11a), which he calls the *i(nternal)-inference* and the *e(xternal)-inference*, respectively. Following Geurts (1999), he observes that they both behave like presuppositions; the reader can check that they indeed do with respect to the tests in (6) and (7). However, given data like (11b), he maintains the view in Heim (1992) and assumes that only the i-inference is determined by the semantic computation, whereas the e-inference is due to pragmatics. We side with Heim and Sudo on this point, and in Section 3, we give an account of the phenomenon that takes well-known criticisms into consideration.

[2] We here talk about presuppositions of a sentence S. A presupposition trigger in S can trigger a presupposition that is filtered or accommodated within S, so that S overall does not carry a presupposition. However, as said above, local accommodation in conditional antecedents is not available with emotive predicates, which are hard triggers (Abrusán 2016). Instead, the state of affairs is somewhat complicated if one considers cases where the emotive predicate occurs in the scope of a negation (perhaps with a special prosodic contour) that can be interpreted metalinguistically (Horn 1985, cases treated as local accommodation under negation in Heim 1983). Crucially, no such process can apply to (10a) because the emotive predicate occurs unembedded.

Similar to the discussion above on emotive predicates, our main observation that motivates deriving the e-inference from the i-inference is the asymmetry between the two in terms of cancellation. Contexts of mistaken belief, which entail the i-inference and deny the e-inference, are sufficient to achieve felicity. Instead, a context of ignorance, which entails the e-inference and denies the i-inference, results in a presupposition failure. This is shown in (12), paralleling (10).

(12) a. Skye mistakenly believes that Eleni fell. They think that Aditi, too, fell.
b. # Skye doesn't know that Eleni fell. They think that Aditi, too, fell.

At this point, we can formulate our two main arguments in favor of the Heim-Sudo hypothesis and against a view that the e-inference is the result of canonical presupposition projection (as in Geurts 1999). First, the infelicity of (12b) is not predicted on any account that assumes that presuppositions generated in the scope of attitude predicates project globally or are accommodated locally. The example shows that whenever the e-inference is present, the i-inference is present, too. Since the relation does not hold in the opposite way, independence of the i-inference from the e-inference is established. In Geurts's DRT-based system, presuppositions project globally out of attitude predicates whenever they are not locally accommodated. If the presupposition is met globally, then additional pragmatic reasoning can return the i-inference. However, pragmatic inferences should be cancelable, and cancellability is the crucial difference between the i- and the e-inference.

Consider (12b) and assume that the presupposition triggered by *too* projects globally: the second sentence should then presuppose that somebody other than Aditi fell. This presupposition is met in the context, because in the first sentence of (12b), *doesn't know that Eleni fell* carries a presupposition that entails that Eleni fell. However, the discourse is infelicitous overall because the i-inference of the second sentence is not met. If this i-inference were the result of additional pragmatic reasoning on top of the semantic computation, one would expect it to be cancelable: (12b) provides the ideal context for this, because it precisely asserts, with its first sentence, that the i-inference does not hold, i.e., that Skye does not think that Eleni fell. However, the i-inference is never cancelable; therefore, there is no evidence that it is computed separately from the e-inference. Even if we were to postulate that this inference is never cancelable, this would remain an *ad hoc* stipulation. Instead, given (12a), it is natural to think that the opposite could hold, and this is what we want to maintain here: the e-inference is cancelable because it is separately derived via pragmatic reasoning, while violation of the lexically-triggered i-inference necessarily leads to infelicity.

The second argument against a view that takes the e-inference to be the result of presupposition projection comes from the parallelism between presupposition triggers under attitudes and (unembedded) emotive predicates. As we have shown above, a unified treatment of the behavior of emotive predicates and presupposition triggers in attitude contexts is very desirable. In both cases, two inferences arise: one is a belief-relative statement and the other is an unmodalized version thereof. Both these inferences project like presuppositions and can be felicitously filtered. In both cases, a context of mistaken belief can cancel the unmodalized presupposition, but a context of ignorance that only satisfies the unmodalized inference does not achieve felicity because it does not meet the belief-relative presupposition. This parallelism strongly calls for a unified treatment of these two cases and a single mechanism that derives one of the two inferences from the other.

However, given the nature of emotive predicates as single lexical items that are both attitude predicates and presupposition trigger, there is no straightforward way to define what a local context different from the global context would be such that one and the same presupposition can either project locally or globally. In this case, the presupposition is triggered by an item in the main clause, and this specific item must trigger all the necessary presuppositions for semantic computation. There is no way the presupposition can optionally project locally or globally in the same sense as with doxastic attitude predicates: if the factive presupposition is not met globally, there is no local context where the presupposition projection could stop. This criticism also applies to versions of DRT where a presupposition can be copied to multiple sites, as proposed by Zeevat (1984).

A qualification is due at this point. The idea that factive predicates have a special syntax, perhaps involving a dedicated factive complementizer, has been around for a while (Kiparsky and Kiparsky 1970, Kratzer 2006, Kastner 2015, i.a.). If the presupposition is triggered by an element that is syntactically distinct from the emotive predicate itself, say the complementizer, then one could say that the emotive predicate, being attitudinal in nature, does indeed provide a local context for the presupposition that is distinct from the global context. This criticism, although warranted, fails to make a distinction between prototypical factive predicates, like *know*, and emotive predicates in terms of presupposition projection. Given that decompositional analyses of factivity always consider *know*, this line of thought would predict that a belief-relative presupposition should be available with *know* too, contrary to fact, as shown in (13) (see also the discussion around examples (8) and (9)).

(13) # Skye mistakenly believes that Aditi likes linguistics. They know that she likes linguistics.

A symmetric argument can be made against the idea, proposed by Karttunen (2016), that the factive inference of emotive predicates is the result of a conventional implicature. Given the standard view (Potts 2004), conventional implicatures are generally associated to specific lexical items, with the notable exception of appositive relative clauses. Although it is possible to think that emotive factives come with the conventional implicature that their complement is true, an equivalent statement can hardly be made for doxastic or bouletic attitude predicates: every attitude predicate should be associated with the conventional implicature that its complement is defined in the world of evaluation, a quite implausible move. Moreover, conventional implicatures, unlike presuppositions, generally introduce new information, whereas the hallmark of the factive inferences of emotive predicates and of e-inferences is that the sentences are felicitous if these propositions are already entailed by the context. Similarly, a view based on conventional implicatures will have a hard time explaining filtering of the factive presupposition or of the e-inference, which stops them from projecting; trying to filter a conventional implicature should instead result in a redundant statement.

In the next section, we will show how the additional inference may arise from the presupposition of the sentence, both in the case of emotive predicates and in the case of presupposition triggers under attitudes, and how this inference ends up behaving like a presupposition. As said in Section 1, we will only rely on a single plausible pragmatic principle, as well as on minimal assumptions about what presuppositions and global accommodation are.

3 A Pragmatic Solution

We will now try and sketch an account of the factive inference of emotive predicates, and of e-inferences in general, in light of the above discussion. Therefore, we will make the assumption that the semantics only generates belief-relative presuppositions for these cases either because they are lexically encoded, as with emotive predicates, or because they arise compositionally as predicted by Satisfaction Theory or dynamic frameworks, for presupposition triggers embedded under attitudes.

In a discussion of i- and e-inferences, Heim (1992) suggests a pragmatic mechanism that builds on an intuition already put forth in Karttunen (1973). Consider again the example in (11a), repeated below. When the i-inference that *Skye thinks that someone else fell* is accommodated, Heim argues, given that accommodated information is taken to be uncontroversial, the hearer derives the e-inference by reasoning that someone else did fall and Skye was in an appropriate position to find out. This position is also shared by Sudo (2014).

(11a) Skye thinks that Aditi, too, fell.
$\rightsquigarrow$ Skye thinks that someone else fell. *(i-inference)*
$\rightsquigarrow$ Someone else fell. *(e-inference)*

A criticism of this view is raised by Geurts (1999) and more recently reinforced by Abrusán (2022:587): "we do not, in typical circumstances, automatically inherit the beliefs of the people we talk about." Geurts (1999:166) additionally argues that generally, the opposite happens in discourse: if a speaker takes something for granted, they normally ascribe the corresponding belief to other people; instead, talking about someone's beliefs (as opposed to knowledge) often implies that those beliefs are not shared by the speaker.

We now sketch an account that precisely builds on this latter intuition. In this way, we preserve a more plausible pragmatic reasoning, but we show that it is nonetheless possible to assume that only belief-relative presuppositions are generated compositionally. To do so, we state the pragmatic principle in (14), which we call *Echochamber*, to encode the pragmatic intuition above, in line with Geurts and Abrusán's criticism. This principle relies on a notion of contexts as sets of worlds and of contextual support as entailment: $c \models p$ iff for all worlds $w \in c$, $p(w) = \mathrm{T}$. For all animate entities x, the modal operator Bel_x returns T of a proposition iff x believes that proposition.

(14) **Echochamber:**
For any context $c_{\langle s,t \rangle}$, given a certain animate entity x_e and proposition $p_{\langle s,t \rangle}$, if $c \models p$, then $c \models \mathrm{Bel}_x\ p$, unless x's ignorance about p is conveyed.

We then need to define how presupposition satisfaction works and what the role of global accommodation is. We assume the framework of Local Satisfaction; future work will hopefully show how the same idea can be implemented in other frameworks where only belief-relative presuppositions can be derived by the compositional semantic module. The definition of presupposition satisfaction in (15) is fairly standard; global accommodation is defined in (16) as the minimal update that has to be made in order to make an utterance felicitous.

(15) **Presupposition satisfaction**
The presupposition of every sub-formula ϕ must be entailed by its local context in order for the whole utterance to be felicitous.

(16) **Global accommodation**
If a sentence presupposing p is uttered in a global context that does not entail p, then minimally update the global context, if possible, so that the presupposition that p can be satisfied *avoiding inconsistent information states*.

The definition of global accommodation in (16) specifies that inconsistent information states have to be avoided. This is trivially observed when the information state in question is the speaker's, which is arguably a subset of the global context. If an inconsistent information state on the side of the speaker were acceptable, global accommodation could rescue utterances in contexts that entail the negation of their presupposition by returning an empty context: empty contexts trivially entail any presupposition. As shown in (17), a sentence cannot be uttered in a context that entails the negation of its presupposition, and global accommodation cannot return an empty context lest the speaker be assigned an inconsistent information state.

(17) *Context:* Skye didn't smoke in the past.
\# Skye stopped smoking.
a. The context c entails $\neg p$;
b. The utterance presupposes p;
c. Global accommodation does not return $c' \models [\neg p \wedge p]$.

It is easy to observe that this affects other individuals' information states, too. Consider the parallel case in (18), where *know* presupposes the truth of its complement. If global accommodation could assign inconsistent beliefs to Aditi, then it could apply in the case below and return a global context where Aditi believes a contradiction, so that the presupposition of the sentence is trivially satisfied. Instead, the utterance is infelicitous, and therefore, global accommodation cannot return inconsistent information states for individuals other than the speaker, either.

(18) *Context:* Aditi believes that Skye doesn't smoke.
\# Eleni knows that Aditi believes that Skye smokes.
a. The context c entails $\mathrm{Bel}_{\mathrm{Aditi}}\ \neg p$;
b. The utterance presupposes $\mathrm{Bel}_{\mathrm{Aditi}}\ p$;
c. Global accommodation does not return $c' \models \mathrm{Bel}_{\mathrm{Aditi}}\ [p \wedge \neg p]$.

Let us now consider how this simple model of presuppositions and global accommodation can derive the behavior of the factive inference of emotive predicates once the principle of Echochamber is assumed. Recall that our hypothesis is that emotive predicates only trigger a belief-relative presupposition, which is presumably encoded lexically. This assumption alone, repeated in (19), is sufficient to explain the presuppositional behavior of this inference and the fact that it can never be canceled: unless it is entailed by the context, the belief-relative presupposition leads to infelicity.

(19) Skye is happy that Aditi likes linguistics.
Only presupposes: Skye believes that Aditi likes linguistics.

Assume now the context in (20) where it is only explicitly stated that Aditi likes linguistics. If Echochamber applies, given that the global context entails that Aditi likes linguistics, it also entails that Skye believes that Aditi likes linguistics. Therefore, the only presupposition that the emotive predicate triggers is satisfied in the context without the need for global accommodation to apply. The presupposition is satisfied by virtue of Echochamber. Of course, nothing changes if the emotive predicate is negated: given that the only presupposition of the sentence is the belief-relative one and that this is satisfied by the context under Echochamber, the sentence is felicitous.

(20) *Context:* Aditi likes linguistics.
Skye is (not) happy that Aditi likes linguistics.
a. The global context c entails $p = [\lambda w\,.\,\text{Aditi likes linguistics in } w]$;
b. By Echochamber, $c \models \text{Bel}_{\text{Skye}}\ p$;
c. The presupposition of the sentence is satisfied.

Because it is a pragmatic assumption, Echochamber can be suspended. As stated in the definition in (14), Echochamber does not apply when someone's ignorance of a proposition is made explicit. Even if more work is needed to determine the exact range of application of this principle, it is necessary that Echochamber must be suspended in mistaken belief contexts and, more generally, whenever an animate entity's doxastic or epistemic state is part of the at-issue content. Otherwise, pragmatics would never allow a speaker to report someone's beliefs when they are different from their own. The suspension of Echochamber in a mistaken belief context is exemplified in (21), where the belief-relative presupposition is satisfied by the mistaken belief context.

(21) *Context:* Skye mistakenly believes that Aditi likes linguistics.
Skye is happy that Aditi likes linguistics.
a. Echochamber is suspended for Skye and $\neg p = [\lambda w\,.\,\text{Aditi doesn't like linguistics in } w]$;
b. The global context c entails $[\neg p \wedge \text{Bel}_{\text{Skye}}\ p]$;
c. The presupposition of the sentence is satisfied.

Instead, a context of ignorance does not satisfy the belief-relative presupposition, which is the only presupposition triggered by the emotive predicate. Consider (22): even if the context entails that Aditi likes linguistics, Skye's ignorance about this fact is made explicit in the context and is sufficient to suspend Echochamber. Given that Skye does not believe that Aditi likes linguistics, the only presupposition triggered by the emotive predicate is not met and the utterance is infelicitous.

(22) *Context:* Skye doesn't know that Aditi likes linguistics.
Skye is happy that Aditi likes linguistics.
a. Echochamber is suspended for Skye and $p = [\lambda w\,.\,\text{Aditi likes linguistics in } w]$;
b. The global context c entails $[p \wedge \neg \text{Bel}_{\text{Skye}}\ p]$;
c. The presupposition of the sentence is not satisfied.

The application of Echochamber to a local context can also explain cases of apparent filtering of a factive presupposition. Consider the case in (23), where the conditional antecedent entails that Aditi likes linguistics. The conditional introduces a local context c_l where the antecedent is true. If Echochamber can affect local contexts, too, then c_l also entails that Skye believes that. Therefore, the presupposition triggered in the consequent of the conditional is satisfied. This part of the account requires Echochamber to apply locally;[3] alternatively, the local effect could be hardwired into the relation that determines the set of accessible worlds of a conditional, and therefore the local context for the consequent. The formalization of this part of the account will benefit from future work that will hopefully also take counterfactual conditionals into consideration.

[3]See Singh (2011) about the locality of another pragmatic principle: *Maximize Presupposition!*.

(23) If Aditi likes linguistics, Skye is happy that she does.
a. The local context c_l entails $p = [\lambda w\,.\,\text{Aditi likes linguistics in } w]$;
b. By Echochamber, $c_l \models \text{Bel}_{\text{Skye}}\ p$;
c. The presupposition of the emotive predicate is locally satisfied.

Finally, we want to account for the fact that hearers infer that the complement of an emotive predicate is true in the absence of a mistaken belief or ignorance context. As discussed above, we crucially maintain that presuppositions are not the inferences that are drawn when sentences containing presupposition triggers are uttered, but rather the conditions that those sentences impose on the context. Instead, the inferences that intuitively come with sentences that contain presupposition triggers are the result of global accommodation, which is a pragmatic operation that updates the global context: the update itself is the inference. The aim is to show that in the case of emotive predicates, the minimal necessary update of certain contexts—those where Echochamber is not suspended—to make the utterance felicitous corresponds to the factive inference.

To see how this works, consider a context where it has not yet been established whether Aditi likes linguistics, as exemplified in (24). In this context, it is both possible that Aditi likes linguistics and possible that she does not. Under Echochamber, both these epistemic possibilities are also ascribed to a salient individual, like Skye. After the utterance that *Skye is happy that Aditi likes linguistics*, consider a hypothetical updated context c^* where global accommodation has only added that *Skye believes that Aditi likes linguistics*. Given that c^* still entails that it is possible that Aditi does not like linguistics, and because Skye's ignorance has not been conveyed as part of the at-issue content, Echochamber is not suspended and still ascribes this possibility to their beliefs: therefore, Skye eventually believes an epistemic contradiction of the form $[p \wedge \Diamond\neg p]$ (Yalcin 2007). This is an inconsistent information state and therefore not a possible outcome of global accommodation. Instead, the minimal update to the context that global accommodation can cause is one where the resulting context c' entails that Aditi likes linguistics and, under Echochamber, that Skye believes so, too. In this updated context, the presupposition of the sentence containing the emotive predicate is satisfied, and the utterance is felicitous. What is more, the factive inference arises secondarily, as a result of global accommodation.

(24) *Context:* It has not yet been established whether Aditi likes linguistics.
Skye is happy that Aditi likes linguistics.
⇝ Aditi likes linguistics.
a. The global context c entails $[\Diamond p \wedge \Diamond\neg p]$ for $p = [\lambda w\,.\,\text{Aditi likes linguistics in } w]$;
b. By Echochamber, $c \models \text{Bel}_{\text{Skye}}\ [\Diamond p \wedge \Diamond\neg p]$;
c. Global accommodation does not return $c^* \models \big[[\Diamond p \wedge \Diamond\neg p] \wedge \text{Bel}_{\text{Skye}}\ [p \wedge \Diamond\neg p]\big]$;
d. Global accommodation under Echochamber returns $c' \models [p \wedge \text{Bel}_{\text{Skye}}\ p]$.

This portion of the account relies on the idea that Echochamber also operates on modalized propositions and in particular, on epistemic possibilities that have not yet been confirmed or ruled out; additionally, epistemic contradictions count as inconsistent information states that are avoided by global accommodation. To see that this is indeed the case, let us consider another example where a predicate has a modalized presupposition. The predicate *make sure* presupposes that the subject believes at a previous time that the complement might be false. This belief-relative presupposition behaves exactly like the cases above, and in out-of-the-blue context, a sentence with *make sure* comes with the inference that this possibility is active for the speaker, too, as shown in (25).

(25) Eleni will make sure that Taro goes to Prague next week.
⇝ Eleni believes that Taro might not go to Prague next week.
⇝ Taro might not go to Prague next week.

Consider the examples in (26) and (27). In a context in which it is settled at the global level that Taro will go to Prague next week (call this proposition p), the sentence that *Eleni will make sure that p* cannot be uttered felicitously because Echochamber has already ascribed to Eleni the belief that Taro will go to Prague next week. Even in this case, global accommodation cannot ascribe an epistemic contradiction to Eleni's information state. Instead, if Echochamber is suspended in a mistaken belief context where Eleni still entertains the possibility that Taro will not go to Prague next week, the sentence can be uttered felicitously because its only presupposition, the belief-relative one, is satisfied globally.

(26) *Context:* Taro will certainly go to Prague next week.
Eleni will make sure that Taro goes to Prague next week.
  a. The global context c entails $p = [\lambda w\,.\,\text{Taro goes to Prague in } w]$;
  b. By Echochamber, $c \models \text{Bel}_{\text{Eleni}}\ p$;
  c. The sentence presupposes $\text{Bel}_{\text{Eleni}} \Diamond\neg p$;
  d. Global accommodation does not return $c' \models \left[p \wedge \text{Bel}_{\text{Eleni}}\ [p \wedge \Diamond\neg p]\right]$

(27) *Context:* Eleni mistakenly believes that Taro might not go to Prague next week.
Eleni will make sure that Taro goes to Prague next week.
  a. Echochamber is suspended for Eleni and $p = [\lambda w\,.\,\text{Taro goes to Prague in } w]$;
  b. The global context c entails $[p \wedge \text{Bel}_{\text{Eleni}}\ \Diamond\neg p]$;
  c. The presupposition of the sentence is satisfied.

Therefore, the case of *make sure* confirms that the framework must care about modalized propositions and that epistemic contradictions are unacceptable (inconsistent) information states for global accommodation. The cases above have covered all the major features of the presupposition-like behavior of the factive inference of emotive predicates: satisfaction, projection, filtering, and accommodation. Finally, the same mechanism applies exactly in the same way to the e-inferences that arise when a presupposition trigger in embedded under an attitude predicate, (11b), thus providing a unified account of the two cases.

4 Conclusions

In this paper, we argued that a careful consideration of the inferences associated with emotive predicates, and a comparison presupposition triggers embedded under attitude predicates, motivates a certain view of what presuppositions are and what the role of global accommodation is; namely one where presuppositions are conditions that sentences impose on the context. Instead, the inferences that are associated to presuppositional sentences are the result of global accommodation. In principle, these inferences do not have to be identical to the presuppositions themselves.

This is the case of emotive predicates or presupposition triggers under attitudes discussed above. These two classes of cases systematically give rise to a belief-relative presupposition and an unmodalized inference (the factive inference and the e-inference, respectively). A unified treatment of these cases is desirable, but it cannot be achieved with the assumption that the unmodalized inference is also a presupposition. Instead, this inference, which is cancelable, systematically and defeasibly arises from pragmatic reasoning.

We sketched an account that tries to meet these desiderata: it applies uniformly to the two families of cases (emotive predicates and presupposition triggers under attitudes), it assumes that only the belief-relative presupposition is an actual presupposition of these sentences, and it derives the presupposition-like behavior of the other inference with the application of a single pragmatic principle, together with a fairly uncontroversial definition of global accommodation. The only innovation about global accommodation is that it cannot return inconsistent information states, but here, we only made explicit what can uncontroversially be observed from cases like (17) and (18).

The pragmatic principle we introduce, which we called Echochamber, follows an intuition already present in the literature, namely that speakers tend to assume that uncontroversial information is shared by other salient individuals they talk about. Once Echochamber is in place, we can derive the behavior of the factive inference that comes with emotive predicates with respect to satisfaction, projection, and filtering. Additionally, we can see a principled reason why that inference is the default result of global accommodation. In contexts of mistaken belief or ignorance, Echochamber is suspended and only the belief-relative presupposition, *qua* semantic presupposition, is active.

Our discussion challenged presupposition analyses that assume default global projection; future work might show how our proposal can be implemented in other frameworks that derive local versions of presuppositions compositionally and in particular, treat attitude predicates as plugs. More work on the exact formalization of our account is also needed, in particular on the application of Echochamber to local contexts, a notion we left informal here. Nonetheless, the point still stands that our view of presuppositions and local accommodation in principle creates the possibility for a unified account of the cases discussed here, a possibility that is not clearly there if presuppositions are inferences and the role of global accommodation is not clearly defined.

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Maik Thalmann
RTG 2636 "Form-Meaning Mismatches"
University of Göttingen
Nikolausberger Weg 23
37073 Göttingen, Germany
maik.thalmann@gmail.com

Andrea Matticchio
Department of Linguistics
UMass Amherst
650 North Pleasant Street
Amherst, MA 01003, USA
amatticchio@umass.edu